MODEL BENCHMARK · BONUS WORK, AFTER THE TIMEBOX

Opus 5.5, Sonnet 5.5 and Haiku 5.5 on two real assessments

5 November 2026 · 12 reports, two runs per model and assessment, through Nettle's normal queue and worker · same prompts, schema and effort (medium) · $2.23 in total, judge included

Set-up

- **Pacific Coast** (Default Insurer): COPE survey in English from 3 photos and 2 notes. Thin evidence, so several of the 22 questions should end as "insufficient evidence".
- **Udine** (Customer A): COPE survey in Italian from the 4-page survey PDF and 3 photos. The engineer's five numbered recommendations in the PDF are the answer key, scored by an Opus judge.

Results (mean of two runs; quotes and retries summed)

Model	Pacific Coast time · cost	Udine time · cost	Answered of 22 (Pacific Coast)	Quotes verified	Expert recs covered (Udine)	Answers rejected and retried
Opus 5.5 (default)	107 s · $0.257	169 s · $0.469	12	244 / 244	5 / 5	0
Sonnet 5.5	52 s · $0.102	87 s · $0.193	10	171 / 171	5 / 5	1
Haiku 5.5	58 s · $0.010	118 s · $0.017	10	152 / 152	5 / 5	2

Findings

- **The mechanics hold for every model.** All 567 quotes were found verbatim in the evidence, and every Udine report covered all five of the expert's recommendations.
- **Cost spans almost 30 times.** Haiku costs under 4% of Opus, Sonnet about 40%. At these prices the engineer's review, not the model, is the expensive part of a survey.
- **Opus finds more in thin evidence:** 12 answers against 10 on Pacific Coast, and 37 quotes per report against 28 (Sonnet) and 23 (Haiku).
- **Haiku overreaches.** In all three of its Pacific Coast runs it rated sprinkler coverage "Partial" from a quote about corrosion; Opus and Sonnet answered "insufficient evidence" in all four of theirs and asked for the sprinkler layout. **A verified quote proves the evidence exists, not that it supports the answer.**
- **Smaller models slip on formats, and the server catches it:** three answers were rejected and retried (an explanation instead of an option; "Moderato" for "Moderate"), each billed and recorded. Opus had none.
- **Speed and cache.** Sonnet was fastest; Haiku thought the most (14k to 27k output tokens per report). A warm prompt cache made Opus's second report 14 to 24% cheaper.

Recommendation

- **Opus 5.5** by default for reports that will be issued: the most cited answers, declines when it should, no retries. It costs $0.15 to $0.30 a report more than Sonnet.
- **Sonnet 5.5** for cost-sensitive customers and drafts: 2.4 times cheaper, twice as fast, same recall, slightly fewer findings.
- **Haiku 5.5** for previews and triage, not for issued reports without a careful review.

Limits and next steps

- Two runs each, so the results show a direction, not statistical proof. The recall test saturates (5 / 5 everywhere) because the advice is part of the evidence: hold it out next time. One Opus judge, not yet calibrated by a human.
- Every run used effort medium: the default for Opus and Haiku, but one level below Sonnet 5.5's default (high). Next: a targeted effort sweep, plus a check that the cited evidence supports each rated answer.

Full write-up: docs/benchmark.md · raw results and the judge's reasoning: docs/benchmark-results.json
Reproduce: `python manage.py benchmark_models` (about $2.50 and 15 minutes with two workers)